UNIT 2: Physical Quantities

Subject: Physics

Grade: Grade 9

Scales, standards, SI units, significant figures, measurement practice, and unit conversion.

Scales and Standards

In physics, a scale is a system used to measure or compare levels of a variable. Four common data scales are nominal, ordinal, interval, and ratio. Physics measurements mostly use ratio scales because they allow meaningful multiplication and division and include a true zero.

Scale	Main Feature	Example
Nominal	Labels only, no numerical ranking	Gender, hair color
Ordinal	Rank order	Class rank
Interval	Equal intervals, no true zero	Temperature in Celsius
Ratio	Equal intervals and true zero	Length, time, mass, temperature in Kelvin

Fundamental SI Quantities and Units

Physics uses internationally accepted standards. Seven basic quantities are used to build all other measurements.

Basic Quantity	Symbol	SI Unit	Unit Symbol
Length	L	meter	m
Mass	m	kilogram	kg
Time	t	second	s
Temperature	T	kelvin	K
Current	I	ampere	A
Amount of substance	n	mole	mol
Luminous intensity	Iv	candela	cd

Scientific Notation and Significant Figures

Scientific notation writes very large or very small numbers in the form $d \times 10^n$, where $1 <= d < 10$ and n is an integer.

Significant figures include:
- all non-zero digits
- zeros between non-zero digits
- trailing zeros in a decimal measurement

Examples:

- 2000 has 1 significant figure
- 2000.0 has 5 significant figures
- 0.000345 has 3 significant figures

Prefixes and Multipliers

Prefixes are attached before unit names to represent powers of ten with no space between prefix and unit.

Prefix	**Symbol**	**Multiplier**	**Exponent**
tera	T	1,000,000,000,000	10^12
giga	G	1,000,000,000	10^9
mega	M	1,000,000	10^6
kilo	k	1,000	10^3
deci	d	0.1	10^-1
centi	c	0.01	10^-2
milli	m	0.001	10^-3
micro	u	0.000001	10^-6
nano	n	0.000000001	10^-9
pico	p	0.000000000001	10^-12

Measurement, Classification, and Conversion

Measurement compares an unknown quantity with a known standard unit. Key tools include ruler/tape for length, beam balance for mass, and stopwatch or clock for time.

Physical quantities are classified as:

- fundamental (directly measured)
- derived (computed from basic quantities)
- scalar (magnitude only)
- vector (magnitude and direction)

Common conversion relations:

Quantity	**Conversion**
Length	1 km = 1000 m, 1 m = 100 cm, 1 m = 1000 mm
Mass	1 kg = 1000 g, 1 g = 1000 mg, 1000 kg = 1 tonne
Time	1 min = 60 s, 1 hr = 60 min, 1 day = 24 hr
Vector vs scalar	distance is scalar, displacement is vector; speed is scalar, velocity is vector

Common Derived Quantities

Derived quantities are calculated from basic quantities.

Physical Quantity	Symbol	Formula	Unit
Speed	v	distance/time	m/s
Density	rho	mass/volume	kg/m^3
Acceleration	a	velocity/time	m/s^2
Force	F	mass x acceleration	N
Work	W	force x distance	J
Pressure	P	force/area	Pa

Practice and Workout Problems

Practice converting and solving measurement tasks:

- Convert 200 m into cm, km, and mm.
- Convert 10 nm into m, mm, and um.
- A beam balance has 1.5 kg, 500 g, 250 g, 25 g, and 0.8 g on one side. Find total mass in g and kg.
- Convert 2 hr, 0.5 hr, and 3/5 hr into seconds.
- Convert 25.5 years into months.
- Convert 8 m 40 cm into cm.
- Find minutes in 3 days.
- If one ceramic tile is 0.25 m^2, how many tiles cover 40 m^2?
- Express 1.5 x 10^11 m using a suitable SI prefix.